MINIMUM SPANNING TREES ON COMPLETE GRAPHS WITH RANDOM EDGE WEIGHTS

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1. Introduction

Consider the complete graph on n vertices where the edge weights are independently and identically distributed as Unif$(0,1)$ random variables. Let W_n be the weight of the minimum spanning tree on this graph. The goal of this paper is to sketch a proof of the following intriguing result about the asymptotic behavior of W_n:

Theorem 1 (Frieze [5] 1985)**.**

$$\lim_{n\to\infty} \mathbb{E}W_n = \zeta(3) \tag{1}$$

where $\zeta(3) = \sum_{k=1}^{\infty} 1/k^3 \approx 1.202$.

Theorem 1 says that the average weight of the minimum spanning tree converges to a constant as the number of vertices approaches infinity. Surprisingly, this constant turns out to be the Riemann zeta function evaluated at 3. Figure 1 shows four realizations of W_n and plots estimates of $\mathbb{E}W_n$ for $n \leq 100$. The plot clearly shows that $\mathbb{E}W_n$ monotonically increases towards $\zeta(3)$ and that the convergence is relatively quick ($\mathbb{E}W_n$ is within 0.01 of $\zeta(3)$ when $n \geq 50$).

Several additional results about W_n are known. In the same paper, Frieze [5] proves that $W_n \to \zeta(3)$ in probability and generalizes Theorem 1 to cases where the edge weights follow non-uniform distributions.

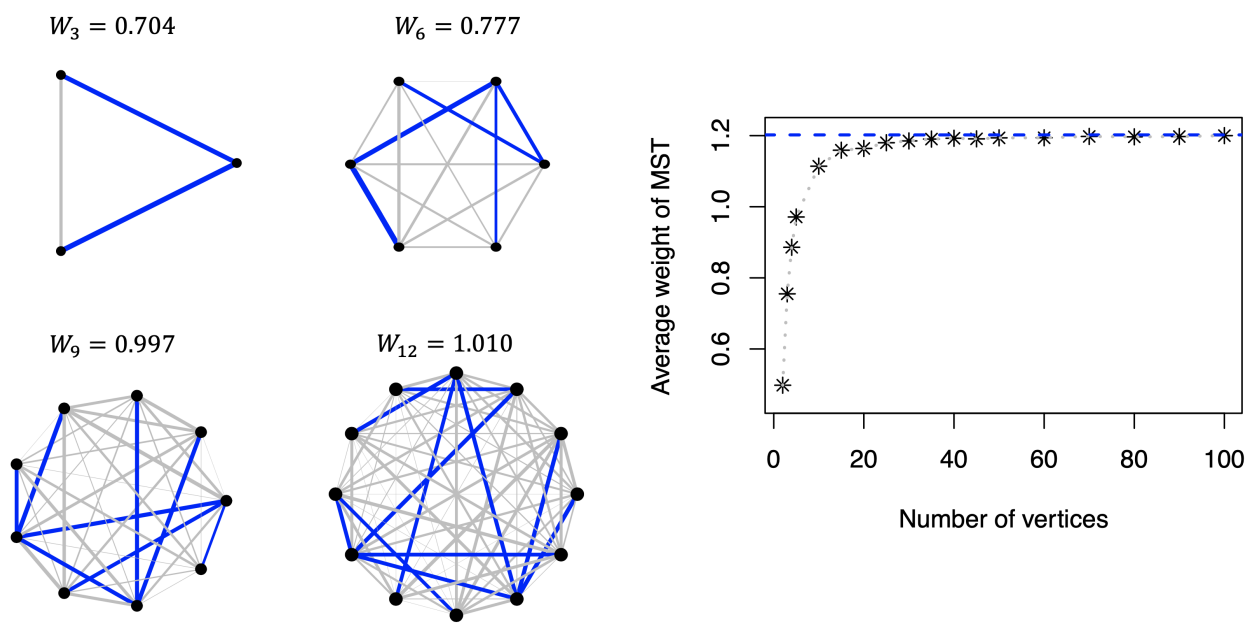

Figure 1. On the left: realizations of W_n for four small values of n. Edge weights are Unif$(0,1)$ random variables and edge thickness is proportional to 1 minus weight. Blue edges are a part of the minimum spanning tree. On the right: estimating $\mathbb{E}W_n$ for various values of $n \leq 100$. $\mathbb{E}W_n$ was estimated by taking the mean of 10^4 trials. The dashed blue line is $\zeta(3)$. Figure made using igraph [2].

Janson [6] proves that W_n is asymptotically normal. Aldous and Steele [1] study a related problem where the vertices are uniformly distributed on $[0,1]^d \subset \mathbb{R}^d$ and the edge weights are the Euclidean distances between vertices.

The main idea behind the proof of Theorem 1 is to use Kruskal's algorithm to build the minimum spanning tree. As we will see, this construction allows $\mathbb{E}W_n$ to be expressed in terms of the number of components in a random graph (which is a well-understood problem). Although our proof sketch closely follows the strategy of Frieze, we add detail where possible and emphasize the connections to the classic ideas in probability and computer science.

2. Kruskal's algorithm

Let $G = (V, E)$ be the complete graph on n vertices with edge weights given by $w : E \to \mathbb{R}$. Suppose that the edges $E = \{e_1, \ldots, e_{\binom{n}{2}}\}$ have been labeled in order of increasing weight, so that $w(e_i) \leq w(e_j)$ if $i \leq j$. Since the edge weights are Unif$(0,1)$ random variables, we have that

$$\mathbb{E}w(e_i) = \int_0^1 x \binom{\binom{n}{2}}{i-1} x^{i-1}(1-x)^{\binom{n}{2}-i} dx = \frac{i}{\binom{n}{2}+1} \tag{2}$$

since the integral in (2) can be seen as the expected value of a Beta $\left(i, \binom{n}{2} - i + 1\right)$ random variable. We use the following algorithm to construct a minimum spanning tree of G:

(1) Initialize $i := 0$, $j := 1$, and F_0 to be G with all edges removed.
(2) If edge e_j connects two distinct components of F_i, define $F_{i+1} := F_i \cup \{e_j\}$ and proceed to step 3. Otherwise, set $j := j + 1$ and repeat step 2.
(3) Set $i := i + 1$. If F_i has $n - 1$ edges, halt and return F_i. Otherwise, set $j := j + 1$ and go back to step 2.

The above procedure was introduced by Kruskal [7] in 1956 and has since become one of the most popular algorithms for computing minimum spanning trees. The correctness of Kruskal's algorithm is immediate from the cut property of minimum spanning trees: for any $S \subset V$ the minimum spanning tree contains the minimum weight edge connecting S to $V - S$. At iteration $i < n - 1$, F_i is a forest (union of disjoint trees), and the edge added to create F_{i+1} is always the minimum weight edge connecting two previously disjoint trees.

Let M_i be the label of the edge that was added to create the forest F_i (for example, if $F_i = F_{i-1} \cup \{e_j\}$ then $M_i = j$). Then (2) implies that

$$\mathbb{E}W_n = \mathbb{E}\left(\mathbb{E}(W_n | M_1, \ldots, M_{n-1})\right) = \frac{1}{\binom{n}{2}+1} \sum_{i=1}^{n-1} \mathbb{E}M_i \tag{3}$$

In particular, we can turn our focus to analyzing $\mathbb{E}M_i$.

3. Components of Erdős-Rényi random graphs

Let G_m denote the graph obtained by restricting to the m edges $e_1, \ldots, e_m$ (ordered by increasing weight). Since the edge weights are independently and identically distributed, G_m can be seen as a draw from the uniform distribution over graphs with n vertices and m edges. This distribution of random graphs is oftentimes called the Erdős-Rényi model due to their seminal work in 1959 [3].

Let $C(G_m)$ be the number of components in G_m. As eluded to earlier, there is a natural connection between $C(G_m)$ and the M_i:

Lemma 2. *If $M_i > m$ then $C(G_m) > n - i$.*

Proof. If $M_i > m$ then G_m has the same number of components as the forest F_k for some $k < i$. The result follows since F_k has $n - k > n - i$ components. □

In light of Lemma 2, we seek bounds on $C(G_m)$ for various m. The first result says that G_m is usually connected when $m = \Omega(n \log n)$:

Lemma 3.

$$\mathbb{P}\left(C\left(G_{2n\log n}\right) > 1\right) = O(n^{-3}) \tag{4}$$

Proof. By the union bound,

$$\mathbb{P}\left(C\left(G_{2n\log n}\right)>1\right) \leq \sum_{S\subset V, |S|\leq n/2} \mathbb{P}\left(G_{2n\log n} \text{ has no edges connecting } S \text{ to } V-S\right) \tag{5}$$

$$= \sum_{k=1}^{n/2} \binom{n}{k} \binom{\binom{n}{2}-k(n-k)}{2n\log n} \Big/ \binom{\binom{n}{2}}{2n\log n} \tag{6}$$

Applying the well-known bound $\left(\frac{r}{s}\right)^s \leq \binom{r}{s} \leq \left(\frac{re}{s}\right)^s$ $(r,s\geq 1)$ to each of the above binomial coefficients, we obtain

$$\sum_{k=1}^{n/2} \binom{n}{k} \binom{\binom{n}{2}-k(n-k)}{2n\log n} \Big/ \binom{\binom{n}{2}}{2n\log n} \leq \sum_{k=1}^{n/2} \frac{(ne)^k \left(\binom{n}{2}-ek(n-k)\right)^{2n\log n}}{k^k \binom{n}{2}^{2n\log n}} \tag{7}$$

Notice that the summands in (7) are decreasing in k. To see this, one can observe that the derivative of $n^k\left(n(n-1)-2ek(n-k)\right)^{2n\log n}$ with respect to k is negative when $1\leq k\leq n/2$. This allows us to bound the sum by n times the first term:

$$\sum_{k=1}^{n/2} \frac{(ne)^k \left(\binom{n}{2}-ek(n-k)\right)^{2n\log n}}{k^k \binom{n}{2}^{2n\log n}} \leq en^2 \frac{\left(\binom{n}{2}-e(n-1)\right)^{2n\log n}}{\binom{n}{2}^{2n\log n}} \tag{8}$$

$$= en^2 \left(\frac{n-e}{n}\right)^{2n\log n} \tag{9}$$

The desired result then follows immediately from the fact that $\left(\frac{n-e}{n}\right)^{2n\log n} \sim n^{-2e}$. □

When m is smaller than $2n\log n$, it is more likely that G_m is comprised of several disconnected pieces. The following (more difficult) result gives us the bound needed in this case:

Theorem 4. *For $x>0$, define the function*

$$f(x) = \frac{1}{2x}\sum_{k=1}^{\infty} \frac{k^{k-2}\left(2xe^{-2x}\right)^k}{k!} \tag{10}$$

Then for $0\leq m\leq 2n\log n$,

$$\mathbb{P}\left(C(G_m) > nf(m/n)+3n^{4/5}\right) = O(n^{-1/6}) \tag{11}$$

To see that the series defining f is convergent, one can use Stirling's approximation on $k^k/k!$ combined with the fact that $xe^{-2x}\leq e^{-1}$ for all $x>0$. Additionally, it is easy to see that f is monotonically decreasing. Since $\lim_{x\to 0^+} f(x)=1$ and $\lim_{x\to\infty} f(x)=0$, this implies that f^{-1} is defined on $(0,1)$ (this fact will be needed later).

The proof of Theorem 4 is long and technical, so we only summarize the main ideas. There are two possibilities when $C(G_m) > nf(m/n)+3n^{4/5}$:

(1) G_m has more than $nf(m/n)+n^{4/5}$ isolated trees of size at most $n^{1/5}$.
(2) G_m has more than $n^{4/5}$ components which are not isolated trees but have at most $n^{1/5}$ vertices.

We proceed by showing that both cases have probability $O(n^{-1/6})$. For the first case, let T be the number of isolated trees of size at most $n^{1/5}$. Then $T=\sum_{k=1}^{n^{1/5}}\tau_k$, where τ_k is the number of isolated trees of size exactly k. The probability that a fixed tree of size k is a component of G_m is

$$\binom{\binom{n-k}{2}}{m-(k-1)} \Big/ \binom{\binom{n}{2}}{m} \tag{12}$$

since $m-(k-1)$ edges must be placed on the $n-k$ vertices that are not a part of the tree. Since there are $\binom{n}{k}k^{k-2}$ possible trees of size k by Cayley's formula, linearity of expectation implies that

$$\mathbb{E}\tau_k = \binom{n}{k}k^{k-2}\binom{\binom{n-k}{2}}{m-(k-1)} \Big/ \binom{\binom{n}{2}}{m} \tag{13}$$

To get an accurate approximation of $\mathbb{E}\tau_k$, we use the bound given in equation 2.13 of [4]:

$$\binom{\binom{n-k}{2}}{m-(k-1)} \Big/ \binom{\binom{n}{2}}{m} = \left(\frac{2m}{n^2}\right)^{k-1} e^{-2mk/n}\left(1+O(m/n^2)\right) \tag{14}$$

See equation 7 of [4] for details on how this bound is obtained. Since $m = o(n^2)$ and $k^2 = o(n)$, applying (14) to $\mathbb{E}\tau_k$ gives

$$\mathbb{E}\tau_k = \frac{n^k}{k!}k^{k-2}\left(\frac{2m}{n^2}\right)^{k-1} e^{-2mk/n}(1+o(1)) \tag{15}$$

$$= n\cdot\frac{n}{2m}\frac{k^{k-2}\left(2(m/n)e^{-2(m/n)}\right)^k}{k!}(1+o(1)) \tag{16}$$

and therefore $\lim_{n\to\infty}\mathbb{E}T = nf(m/n)$. Applying Chebyshev's inequality, we have

$$\mathbb{P}(T - nf(m/n) \geq n^{4/5}) \leq \mathbb{P}(T - \mathbb{E}T \geq n^{4/5}) \tag{17}$$

$$\leq \frac{\mathrm{Var}(T)}{\mathrm{Var}(T)+n^{8/5}} \tag{18}$$

Using similar techniques (and more precise bounds), Frieze proves that $\mathrm{Var}(T) = O(n^{7/5}\log n)$ (Lemma 4 of [5]). Applying this bound to (18) gives the desired $O(n^{-1/6})$ result.

For the second case, let S be the number of components of G_m which are not isolated trees but have at most $n^{1/5}$ vertices. A component of size k with ℓ edges can be constructed by first choosing a tree and then adding $\ell-(k-1)$ edges. This gives $\binom{n}{k}k^{k-2}\binom{k}{2}^{\ell-(k-1)}$ as an upper bound on the number of such components. By the same technique used in the previous case, we have

$$\mathbb{E}S \leq \sum_{k=1}^{n^{1/5}}\sum_{\ell=k}^{\binom{k}{2}}\binom{n}{k}k^{k-2}\binom{k}{2}^{\ell-(k-1)}\binom{\binom{n-k}{2}}{m-\ell}\Big/\binom{\binom{n}{2}}{m} \tag{19}$$

Applying the bound in (14) again, we obtain

$$\mathbb{E}S \leq \sum_{k=1}^{n^{1/5}}\sum_{\ell=k}^{\binom{k}{2}}\frac{n^k k^{k-2}}{k!}\left(\frac{k(k-1)}{2}\right)^{\ell-(k-1)}\left(\frac{2m}{n^2}\right)^{\ell} e^{-2mk/n}(1+o(1)) \tag{20}$$

(21)

Using the fact that $k^2 = o(n)$, we clear the n^k and the $(k(k-1))^{\ell-(k-1)}$ from the numerator:

$$\mathbb{E}S \leq \sum_{k=1}^{n^{1/5}} k^2 \sum_{\ell=k}^{\binom{k}{2}}\frac{1}{2^{\ell-(k-1)}}\frac{k^k}{k!}\frac{2m^\ell}{n^{\ell+1}}e^{-2mk/n}(1+o(1)) \tag{22}$$

$$\leq \sum_{k=1}^{n^{1/5}} k^2 \sum_{\ell=k}^{\binom{k}{2}}\frac{1}{2^{\ell-(k-1)}}\frac{(2n\log n)^\ell}{n^{\ell+1}}e^{-2mk/n+k}(1+o(1)) \tag{23}$$

$$\leq \sum_{k=1}^{n^{1/5}} k^2 \sum_{\ell=k}^{\binom{k}{2}}\frac{1}{2^{\ell-(k-1)}}\cdot O(1)\cdot e^{-2mk/n+k}(1+o(1)) \tag{24}$$

In the case $m > n/2$, $e^{-2mk/n+k} = O(1)$.[1] Then

$$\mathbb{E}S \leq \sum_{k=1}^{n^{1/5}} k^2 \sum_{\ell=k}^{\binom{k}{2}}\frac{O(1)}{2^{\ell-(k-1)}} = \sum_{k=1}^{n^{1/5}} O(k^2) = O(n^{3/5}) \tag{25}$$

Applying Markov's inequality, we have

$$\mathbb{P}(S > n^{4/5}) \leq n^{-4/5}\mathbb{E}S = O(n^{-1/5}) = O(n^{-1/6}) \tag{26}$$

which concludes the proof of Theorem 4.

[1] One can handle the case where $m = o(n)$ separately, see the first paragraph on page 54 of [5].

4. Proof of Theorem 1

We now use the earlier results to prove that $\limsup_{n\to\infty} \mathbb{E}W_n \le \zeta(3)$. For $i \le n - 3n^{4/5}$, set $b_i = f^{-1}(1 - i/n - 3n^{-1/5})$ and notice that

$$\mathbb{E}M_i \le nb_i + 2n\log n\mathbb{P}(nb_i < M_i \le 2n\log n) + \binom{n}{2}\mathbb{P}(M_i > 2n\log n) \tag{27}$$

If $M_i \ge 2n\log n$, then Lemma 2 implies that $C(G_{2n\log n}) > n - i > 1$. From this, Lemma 3 gives that $\mathbb{P}(M_i \ge 2n\log n) = O(n^{-3})$. Similarly, if $M_i > nb_i$ then $C(G_{nb_i}) > n - i$. Since

$$n - i = nf(f^{-1}(1 - i/n - 3n^{-1/5})) + 3n^{4/5} = nf(b_i) + 3n^{4/5} \tag{28}$$

Theorem 4 can be applied to conclude that $\mathbb{P}(nb_i < M_i < 2n\log n) = O(n^{-1/6})$. Putting this all together, we get

$$\mathbb{E}M_i \le nb_i + O(n^{5/6}\log n) + O(n^{-1}) \tag{29}$$

If instead $i > n - 3n^{4/5}$, the weaker bound

$$\mathbb{E}M_i \le 2n\log n + \binom{n}{2}\mathbb{P}(M_i > 2n\log n) = O(n\log n) \tag{30}$$

will suffice for our purposes. Returning to the expression for $\mathbb{E}W_n$ found in (3), we obtain

$$\mathbb{E}W_n = \frac{\sum_{i=1}^n \mathbb{E}M_i}{\binom{n}{2}+1} \le \frac{\sum_{i=1}^{n-3n^{4/5}} \left(nb_i + O(n^{5/6}\log n)\right)}{\binom{n}{2}+1} + \frac{O(n^{9/5}\log n)}{\binom{n}{2}+1} \tag{31}$$

$$= \frac{\sum_{i=1}^{n-3n^{4/5}} nb_i}{\binom{n}{2}+1} + o(1) \tag{32}$$

Taking the limit as $n \to \infty$ gives

$$\limsup_{n\to\infty} \mathbb{E}W_n \le \lim_{n\to\infty} \frac{n\sum_{i=1}^{n-3n^{4/5}} f^{-1}(1 - i/n - 3n^{-1/5})}{\binom{n}{2}+1} \tag{33}$$

$$\le \lim_{n\to\infty} \frac{n^2\sum_{i=1}^{n} n^{-1}f^{-1}(1 - i/n)}{\binom{n}{2}+1} \tag{34}$$

$$\le 2\int_0^1 f^{-1}(1-x)dx \tag{35}$$

Since f^{-1} is defined on $(0,1)$, it follows by symmetry that $\int_0^1 f^{-1}(1-x)dx = \int_0^\infty f(x)dx$. Then the desired result is obtained by computing the integral

$$2\int_0^\infty f(x)dx = 2\int_0^\infty \frac{1}{2x}\sum_{k=1}^\infty \frac{k^{k-2}\left(2xe^{-2x}\right)^k}{k!}dx \tag{36}$$

$$= \sum_{k=1}^\infty \frac{2k^{k-2}}{k!}\int_0^\infty (2x)^{k-1}e^{-2xk}dx \tag{37}$$

The inside integral can be evaluated in terms of the gamma function:

$$\int_0^\infty (2x)^{k-1}e^{-2xk}dx = \frac{1}{k^{k-1}}\int_0^\infty (2xk)^{k-1}e^{-2xk}dx \tag{38}$$

$$= \frac{(k-1)!}{2k^k} \tag{39}$$

Plugging (39) back into (37), we conclude that

$$\limsup_{n\to\infty} \mathbb{E}W_n \le \sum_{k=1}^\infty \frac{2k^{k-2}(k-1)!}{2k^k k!} = \sum_{k=1}^\infty \frac{1}{k^3} = \zeta(3) \tag{40}$$

The final step in the proof of Theorem 1 is showing that $\liminf_{n\to\infty} \mathbb{E}W_n \ge \zeta(3)$. We do not include the calculation here since the general idea is exactly the same as the above. The only additional tool needed to

compute the limit inferior is the analogue of Theorem 4 that gives a lower bound for $C(G_m)$ (see Lemma 1 of [5]).

References

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